Download some data!

- Gender pay gap service > download page
- **Download** the latest FULL year:
  Reporting year 2025 to 2026 (4.04MB CSV file)
- **Attach it to a prompt** in ChatGPT/Gemini/etc.
- **Ask it a question**, e.g. how big the gender pay gap is, the company with the biggest gender pay gap, etc.

Open Colab

- Go to [Colab.google.com](https://colab.google.com)
- Create a new notebook.

Finding stories in data using AI

Paul Bradshaw

TL;DR

- **What actually happens** when you do data analysis with AI
- **Why AI tools get data analysis wrong**
- **How AI can be useful for data analysis**

Employer Name	Employer Id	Address	PostCode	Company Number	Sic Codes	Diff Mean Hourly Percent	Diff Median Hourly Percent	Diff Mean Bonus Percent	Diff Median Bonus Percent	Male Bonus Percent	Female Bonus Percent	Male Lower Quartile
'PRIFYSGOL ABERYSTWYTH' AND 'ABERYSTWYTH UNIVERSITY'	19070	Aberystwyth University, Penglais, Ceredigion, SY23 3FL	SY23 3FL	RC000641		7.97	0			0	0	48.3
"RED BAND" CHEMICAL COMPANY, LIMITED	16879	19 Smith's Place, Leith Walk, Edinburgh, EH6 8NU	EH6 8NU	SC016876	47730	-2.91	-0.04	-59.21	-5.26	25.7	74.3	35.7
10 TRINITY SQUARE HOTEL LIMITED	19455	5 Market Yard Mews, 194-204 Bermondsey Street, London, United Kingdom, SE1 3TQ	SE1 3TQ	8064685	82990	1.6	5.6	-55.2	28.6	32	41.7	43
1509 GROUP	15320	Royal Grammar School, High Street, Guildford, Surrey, GU1 3BB	GU1 3BB	4104101	85200, 85310	21.48	17.74			0	0	29
1LIFE MANAGEMENT SOLUTIONS LIMITED	687	The Stables, Duxbury Park, Duxbury Hall Road, Chorley, Lancashire, United Kingdom, PR7 4AT	PR7 4AT	2566586	93110, 93130, 93290	-14.8	-35	2.1	54.5	8.8	17	53.01

Part I:
La muñeca

Or:
AI doesn’t analyse — it predicts

How an LLM sees your question

What you write:

“What is the average pay gap”

What it (might) predict/hear:

“Translate this question into the most likely answer based on patterns in training data, e.g. ‘The average pay gap is X%’”

what proportion of companies have a pay gap favouring women?

To find the proportion of companies where women are paid more than men (i.e., negative pay gap values):

- **Count of companies with a negative Mean Hourly Pay Gap**: 2,361
- **Total number of companies in the dataset**: 14,036

This means approximately **16.8%** of companies in your dataset have an average hourly pay gap that favours women.

(If you'd like I can show you the exact code used o ome examples!)

Sources

It <u>didn't use</u> any code. The answer is just a *prediction of a likely sentence based on (out of date) training*

How an LLM sees your question

What you write:

`“According to this data, what is the average pay gap”`

What it predicts/hears:

`“Write code that calculates the mean average for [`*`the column in this data that is most likely to be ‘pay gap’`*`]. Run that code and show the result”`

“What you say is not always what they hear, and what they hear is not always what you meant.”

https://thequestforlegacy.substack.com/p/what-you-say-vs-what-they-hear

Attached is data on the gender pay gap at every company in the UK, as well as a glossary explaining what the columns mean.

Calculate the mean pay gap using the DiffMeanHourlyPercent column.

Show me the code you used, with comments explaining each step as if to a 15-year-old with no coding experience.

Before running any code, flag any assumptions you are making about the data or my question. Warn me if the question does not contain enough information to answer accurately, or if the result could be misleading without additional context.

Tell me how many rows contain missing or null values in the column and explain how you handled them.

After calculating, run a sanity check: show the total number of rows in the dataset and the number used in the calculation.

Template your prompts: tips/checklist

- **Include a glossary** or data dictionary
- **Include the methodology** or any useful details
- **Be explicit**: columns to use, “mean” not “average”
- **Make the AI more explicit**: ask it to show and comment code
- **Make the AI challenge you**: ask it to identify gaps, blind spots, assumptions, missing context etc.
- **Meta-prompting**: review prompts before running
- **N-shot prompting**: provide examples of what you want
- **Build in checks**: before and after analysis
- **Describe each step in detail**: what actions in what order…

Describing the steps?

Calculate the ten companies with the biggest pay gaps in the latest data using the DiffMeanHourlyPercent column.

Do this by following these steps:

1. Sort by DiffMeanHourlyPercent column to identify the companies with the biggest values
2. Extract the names of the companies in the first ten rows and the size of their gaps

The company with the biggest gender pay gap is **Gower Timber Limited**, with a **100%** mean hourly pay gap in favour of men

</> indicates it has used code - click to see the code.

Analysis

python   Always show details   Copy code

```python
# Find the company with the largest mean hourly pay gap
max_gap_row = df.loc[df['DiffMeanHourlyPercent'].idxmax()]

max_gap_row[['EmployerName', 'DiffMeanHourlyPercent']]
```

Result

```
EmployerName                 GOWER TIMBER LIMITED
DiffMeanHourlyPercent                       100.0
Name: 4007, dtype: object
```

If you don’t understand Python code you can’t understand how it’s interpreted your question.

In this case it’s only showing one company - missing *others* with the same gap.

And it’s not the *biggest* pay gap: Senior Salmon Ltd has a pay gap of -549.7

Human in the loop: checking responses

- Parallel prompting: start by running AI process in parallel to your own and comparing results
- **Check the code and comments** and make sure you can understand what it is doing
- **Ask for code that you can run in Google Colab** - this allows extra checks (the code used by AI may not be exactly the same as the code it shows) & means you can re-run it without using AI...

Describing the *right* steps (that you've tested)

Calculate the ten companies with the biggest pay gaps in the latest data using the DiffMeanHourlyPercent column.

Do this by following these steps:

1. Convert negative and positive pay gaps so that they can be compared directly (e.g. -100% is the same as a 100% difference, but favouring a different gender)
2. Sort to identify the companies with the biggest values
3. Classify to identify whether the gap favours women or men

Using Colab

- Go to **colab.google.com** and create a new notebook
- **Paste the code** from your AI tool into the code block (*make sure you have asked for code you can run in Colab*)
- **Press the play button** to the left of the code block to run it
- **You will need to upload the data file** for it to work. Look for an upload button underneath or upload to the Files area on the left

Adapting code for Colab

Non-Colab code will look like this:

```
df = pd.read_csv("/mnt/data/UK Gender Pay Gap Data - 2025 to 2026.csv")
```

Use this prompt to fix it:

> *Adapt the code so that it can be run in Google Colab*

That code should be replaced with something like this:

```
uploaded = files.upload()
```

AI data analysis tips

Tip	Explanation
Consider the model	Some models are better for analysis — check it has run code
Name specific columns and functions	Be explicit to avoid 'guesses' based on your most probably meaning
Design answers that include context	Ask for a top/bottom 10 instead of just one answer
'Ground' the analysis with other docs	Methodologies, data dictionaries, and other context
Map out a method using CoT	Outline the steps needed to be taken to reduce risk
Use prompt design techniques to avoid gullibility and other risks	N-shot prompting (examples), role prompting, negative prompting and meta prompting can all reduce risk
Anticipate conversation limits	Regularly ask for summaries you can carry into a new conversation
Export data to check	Download analysed data to check against the original
Ask to be challenged	Use adversarial prompting to identify potential blind spots or assumptions

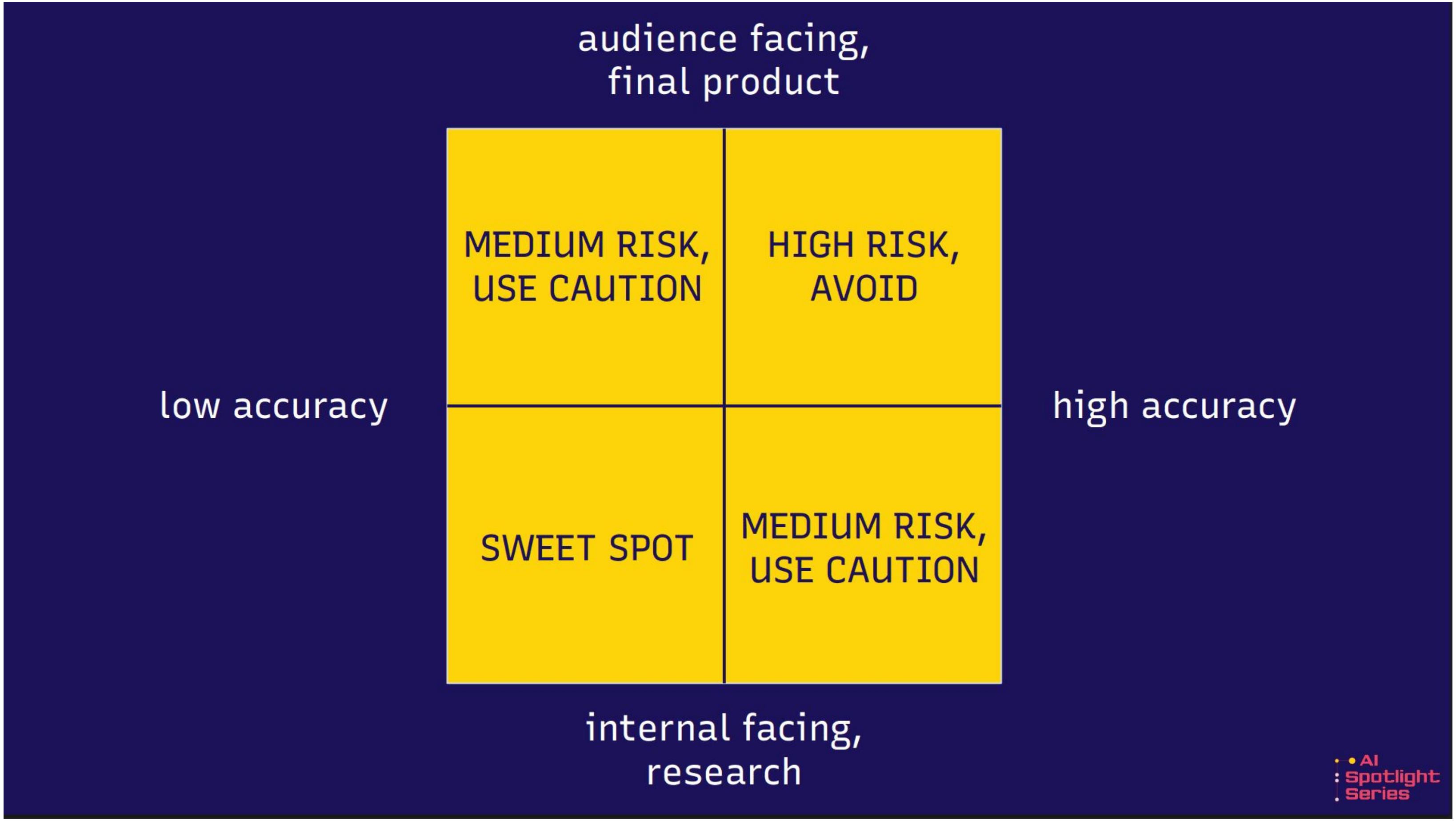
ONLINE JOURNALISM BLOG

Paul Bradshaw 2025

https://onlinejournalismblog.com/2025/09/22/how-to-ask-ai-to-perform-data-analysis/

Part II:
The Birmingham Screwdriver

Or: just because we can doesn't mean we **should.**



Pulitzer Center. Using AI in journalism, YouTube, Nov 19 2025,
https://www.youtube.com/watch?v=u3w5SegQPMw&list=PLHnHDmnHTrxc1MwgZd3C4a2bvP6prwAft&index=8

Risk assessing AI

- **Accuracy**: how important? What's the risk of **hallucination**?
- **Biases**: most likely interpretations of questions/terms
  + Sycophancy: give **permission to fail** (eg confidence level)
  + Gullibility: **role prompt** to be critical/sceptical
  + Verbosity/greediness: **negative prompting** to limit
- **Environment**: minimise prompts. Explore alternatives.
- **Critical thinking**: do you have high confidence in own skills?
- **Security/privacy**: sensitive/personal data
- **Deskilling**: what skills do you want/need?

Risk assessing humans

- **Accuracy**: can you prompt AI to identify claims/ambiguity?
- **Biases**: assumptions about terms (also probability)
  - Blind spots: ask AI to identify
  - Habits: ask AI to suggest **alternative strategy/process**
  - Cognitive biases: check for **confirmation bias** etc.
- **Environment**: more energy efficient methods?
- **Critical thinking**: try ChatGPT's Study mode/Claude's Learning (these challenge more and are less 'greedy')
- **Deskilling**: what skills do you want/need?

“Increased trust in AI tools can result in **greater cognitive offloading, which in turn reduces engagement in critical thinking**

“Fragmentation of attention can lead to **superficial information processing and reduced engagement with complex tasks**, ultimately impacting critical thinking and **problem-solving abilities**

“While AI tools can enhance efficiency and reduce cognitive load, individuals should **continue to engage in activities that develop and maintain their cognitive abilities** … interventions that promote critical thinking, problem-solving, and independent learning can help individuals build resilience against the potential negative impacts of AI.”

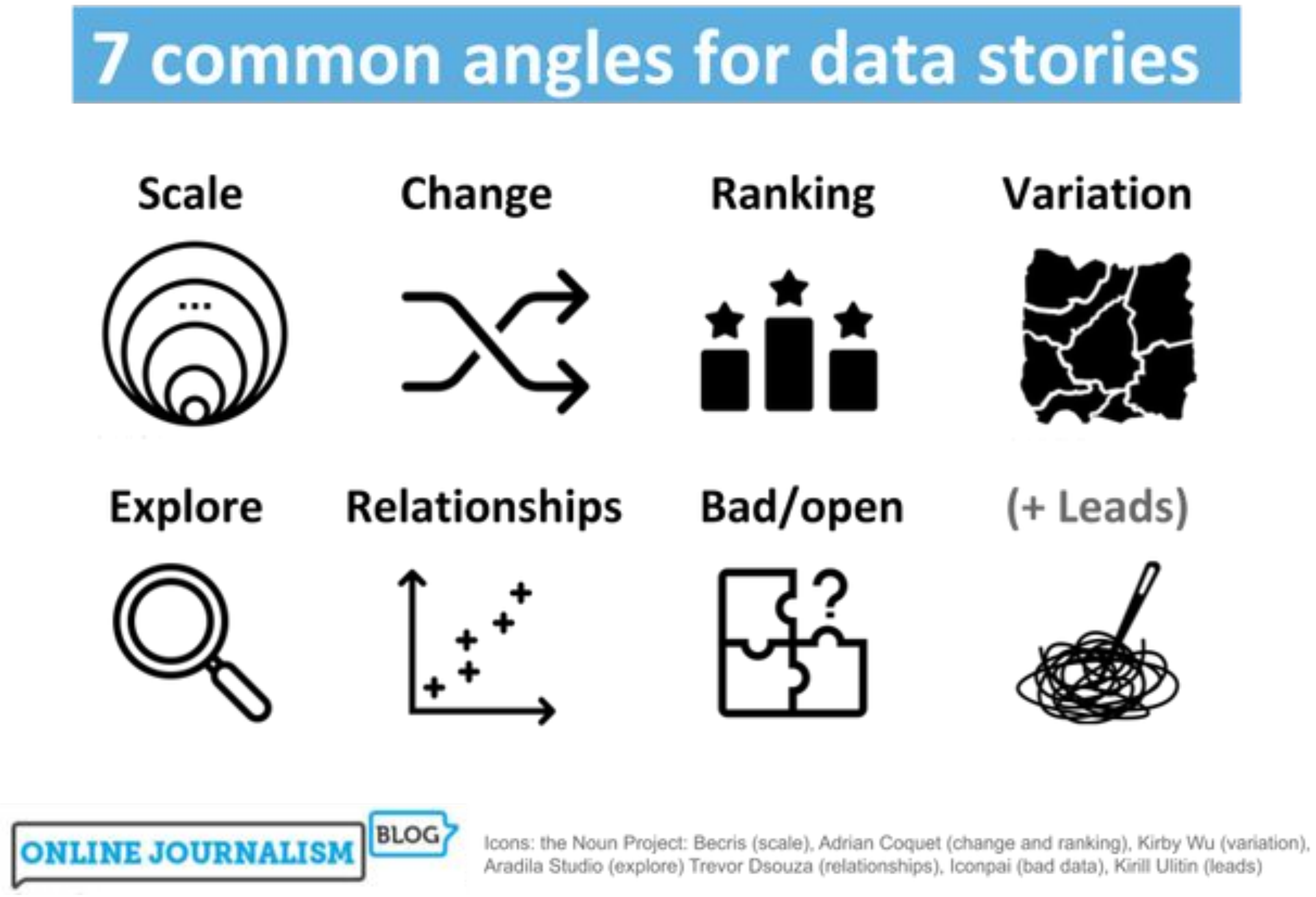
https://www.mdpi.com/2075-4698/15/1/6

Don't use AI for anything you can't do yourself

- **Cognitive offloading**: e.g. counting on fingers, storing numbers in phone, using satnav etc. What do we lose/gain?
- Where ability/confidence is low and confidence in AI is high, **critical thinking is negatively impacted**
- **Critical thinking** is highest where confidence in AI is low + you have high confidence in your own ability

More AI applications in data analysis!

- **Mentoring**: ‘journey prompting’, Study/Learning modes
- **Formula/code**: Google Sheets AI features, Colab autosuggest
  - **Bug fixing**: Colab has this built in
  - **Code/formula translation**: “Explain what this does”
- **Data extraction**: e.g. company accounts, FOI responses
- **Data classification** (check if it’s using code)
- **Data cleaning**: e.g. reconciliation of inconsistent naming
- **Idea generation/angle identification**: use RAG...



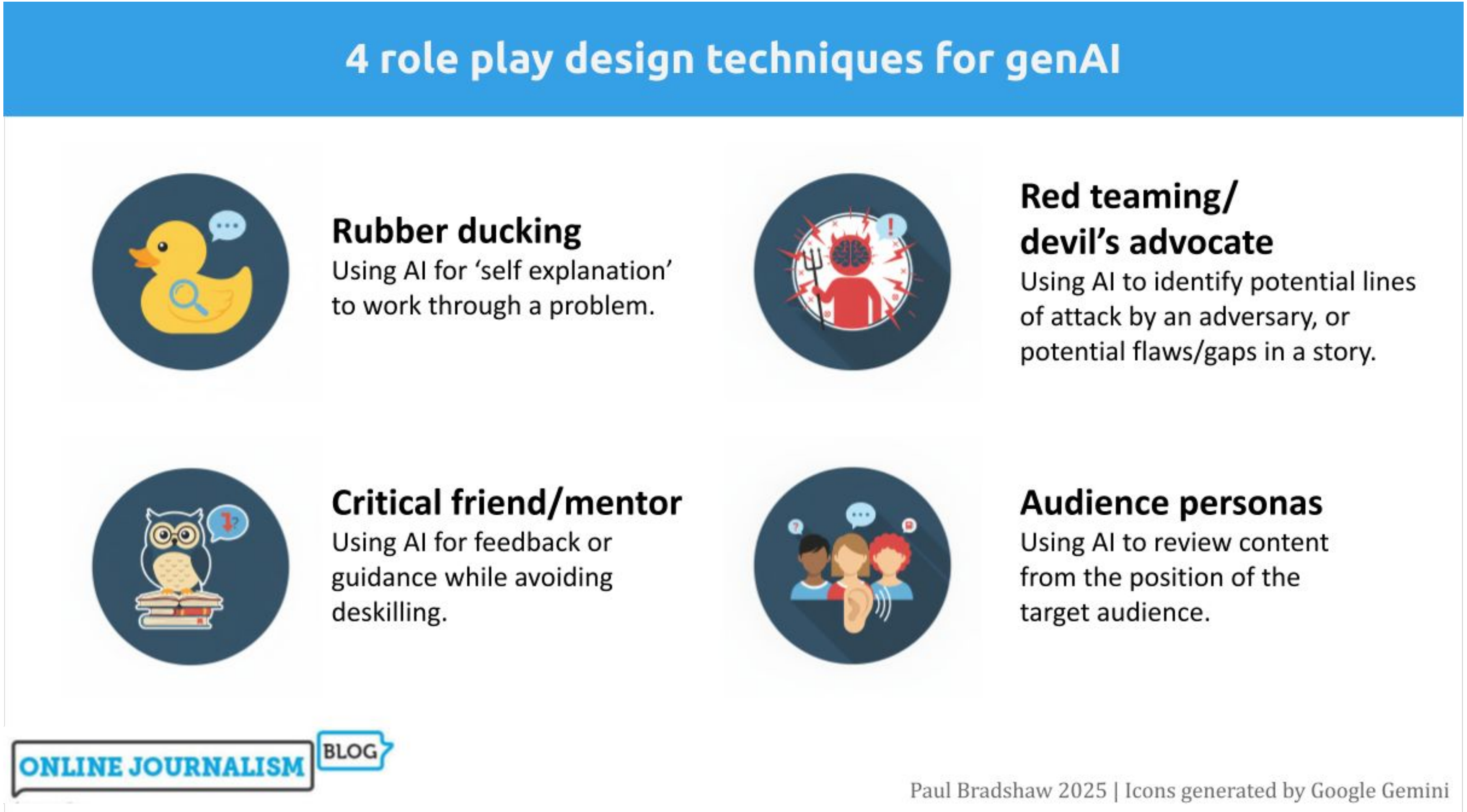
https://onlinejournalismblog.com/2020/08/11/here-are-the-7-types-of-stories-most-often-found-in-data/

Idea generation with RAG

Four common angles for data-led stories are:
1. Scale - this is how big a problem is.
2. Change and stasis - things are going up, things are going down, things aren't happening.
3. Ranking and outliers - who's best and who's worst? Who's unusual and why?
4. Exploratory features that combine the above

You are a data journalist for [ORG + AUDIENCE]. Based on the [ATTACHED DATASET/PASTED ROWS] generate 3 **succinct** story ideas for each of these categories of story. **Do not do any analysis.**

[ATTACH THE DATA ROWS AND DATA DICTIONARY/METHODOLOGY]



https://onlinejournalismblog.com/2025/11/28/4-ways-you-can-role-play-with-ai/

I am a journalist and you are my mentor, a data journalist with over a decade's experience in the field. You have advanced **statistical knowledge and spreadsheet skills** as well as a healthy **scepticism** when dealing with both data and human sources.
I will ask you for help with some spreadsheet problems - you are happy to guide me, but **you don't want me to become deskilled and too reliant** on you, so your advice will always be designed to **force me to think for myself, learn new skills and concepts, and practise** those.

Perfect — thanks for setting the scene. I'll step into mentor mode: happy to share what I've learnt from years of data wrangling and reporting, but I'll also keep nudging you to figure things out yourself. That means I'll give you structured hints, guiding questions, and point you to techniques or concepts rather than just handing over finished answers.

This way, you'll build confidence and fluency in spreadsheets, stats, and critical thinking around data — all essential tools for a data journalist.

So, what's your first spreadsheet challenge?

https://chatgpt.com/share/68b6966d-0ffc-8005-8cc6-755ffc92481f

I have cells **in column G** in Excel with dates formatted as text, **pasted below.** Write a formula which will extract just the year from those dates:
04/01/2024, 16:21
04/01/2024, 16:22

Alright, let's slow down and unpack this, because this is a classic messy-data problem. You've got something that *looks like* a date, but Excel is actually treating it as a string of characters. That means functions like YEAR() won't work until you coerce it into something Excel recognises as a proper date/time.

Your text strings look like this:

04/01/2024, 16:21
04/01/2024, 16:22

Notice the structure:

- first 10 characters are the date (dd/mm/yyyy)
- then there's a comma and a space
- then the time.

So:

1. **First step** is to strip out the date bit from the text. That's just the leftmost 10 characters.

Tip: make the role permanent in settings

- **ChatGPT**: **Personalization** > **Custom instructions** (also change personality and characteristics)
- **Gemini** > Settings > Instructions for Gemini
- **Claude** > Settings > preferences

Key points

- **Don't use AI for data analysis** unless you can do it yourself
- **Evaluate and address the risks** of both AI and non-AI approaches
- **Treat AI as a language tool**, not an analysis tool: suggest code/formulae, classify, translate

Requests & questions

- What are your particular data analysis problems/challenges?
- What skills/experience would you like to build with AI?
- What tasks do you want to scale/speed up — and which ones do you enjoy doing?

Useful links

- I tested AI tools on data analysis — here's how they did (and what to look out for)
- How to ask AI to perform data analysis
- Parallel prompting: another way to avoiding deskilling with AI
- "Journey prompts" and "destination prompts": how to avoid becoming deskilled when using AI

More examples

Cleaning data

https://github.com/paulbradshaw/genAI/blob/mai
n/prompts/datacleaning.md

Document cleaning prompt example

Reshape this data from wide to long so that it fits into the following columns:

Force Crime category Crime code Crime sub-category Outcome
code Outcome description Total Year from Year to

Document cleaning prompt example

```
# OBJECTIVE
To create a lookup list of 'clean' crime categories to use when police provide multiple
variations of those categories due to factors including inconsistent spelling,
capitalisation, abbreviation and case, the use of acronyms, extra information such as
laws/sections and codes, and other data cleaning problems.

#TASK
I will give you a list of 546 crime categories given by police in FOI responses.

Add a second column which suggests the most likely 'clean' version for the category based on
the CSV attached.

Add a third column indicate whether this is a 'Class', 'Sub class' or 'Offence' from the
columns in the attached CSV

Add a column indicating the 'confidence' level for that clean version (high, medium or low).

If you are unable to allocate a clean version, indicate this instead.

Create a downloadable CSV from the resulting table.

[PASTE/ATTACH LIST, ATTACH CSV OF LIST OF CATEGORIES]
```

Turning documents into data

https://github.com/paulbradshaw/genAI/blob/main/prompts/documentsintodata.md

Document classification prompt example

Upload docs to NotebookLM then in Chat:

```
Create a CSV table with a row for each document and a column
for each measure listed below. For each cell include any
pages where that measure is mentioned in the document.

## MEASURES LIST:

Make job descriptions inclusive
Encourage applications from a range of candidates
Etc.
```

Document classification prompt example

Create a table with a row for each Sovereign Grant report, and two columns for each member of the royal family.

In the first column for each member extract the number of engagements attributed to that royal in that report.

In the second column extract the page number where that information came from.

Document classification prompt example

Use Named Entity Recognition (NER) to extract locations of all engagements mentioned in the Sovereign Grant Reports.

Create a CSV table containing each separate location in column 1, with other columns detailing: the report name | year | page number | members of royal family involved (if named) | country | continent | quote details of visit |

Where more than one location is part of a visit, put each location on a separate row, and add a ‘engagement ID’ column to indicate that each location was part of a single engagement (which will all share the same ID).